Zhenyao Cui

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Research Interests

Transferable EEG decoding for practical brain–computer interfaces, with an emphasis on robust generalization across users, tasks, and deployment settings.

Education

Huazhong University of Science and Technology Sep 2024 – Jun 2027

M.Eng. in Intelligent Science and Technology (expected)

- GPA **3.89/4.00**, rank **5/65**; selected for direct graduate admission through the national recommendation scheme.
- Advisor: Prof. Dongrui Wu, IEEE Fellow and Editor-in-Chief of *IEEE Transactions on Fuzzy Systems*.

Huazhong University of Science and Technology Sep 2020 – Jun 2024

B.Eng. in Automation, Elite Engineer Program

- GPA **3.91/4.00**, rank **3/28**; Outstanding Graduate of HUST.

Publications

† equal contribution.

Published and Accepted

1. **Zhenyao Cui**, Siyang Li, Siyuan Kan, Yudi Wang, Fei Luo, and Dongrui Wu. "A Fuzzy Set Based Classification-to-Regression Extension Framework for Transfer Learning and Its Application to Brain-Computer Interfaces." *IEEE Transactions on Emerging Topics in Computational Intelligence*, Early Access, 2026.
   doi:10.1109/TETCI.2026.3683713 | code
2. Siyang Li†, Jiayi Ouyang†, **Zhenyao Cui**†, Ziwei Wang, Tianwang Jia, Feng Wan, and Dongrui Wu. "Backpropagation-Free Test-Time Adaptation for Lightweight EEG-Based Brain-Computer Interfaces." *IEEE Journal of Biomedical and Health Informatics*, accepted for publication, 2026.
   arXiv:2601.07556
3. Dingkun Liu, Yuheng Chen, Zhu Chen, **Zhenyao Cui**, Yaozhi Wen, Jiayu An, Jingwei Luo, and Dongrui Wu. "EEG Foundation Models: Progresses, Benchmarking, and Open Problems." *National Science Review*, accepted for publication, 2026.
   arXiv:2601.17883
4. Siyuan Kan, Huanyu Wu, **Zhenyao Cui**, Fan Huang, Xiaolong Xu, and Dongrui Wu. "CMCRD: Cross-Modal Contrastive Representation Distillation for Emotion Recognition." *IEEE Transactions on Affective Computing*, 17(2):1837–1846, 2026.
   doi:10.1109/TAFFC.2026.3664341

Under Review

5. **Zhenyao Cui**, Jiahui Chai, Siyuan Kan, Yudi Wang, and Dongrui Wu. "MCL-Align: Marginal–Conditional–Label Alignment for Cross-Subject EEG Regression." Under review, *IEEE Transactions on Biomedical Engineering*, 2026.
6. **Zhenyao Cui**, Siyuan Kan, Siyang Li, Ziwei Wang, and Dongrui Wu. "SCORE: Subject Coordinate Recovery for Label-Free Cross-Subject EEG-to-Image Retrieval." Manuscript under review, 2026.
7. **Zhenyao Cui**, Siyuan Kan, Dingkun Liu, and Dongrui Wu. "Beyond Trial Averaging: Anchoring Neural and Visual Representations for Few-Repetition Brain-to-Image Retrieval." Manuscript under review, 2026.

Research Experience

Transferable EEG Decoding Across Users and Deployment Settings
Apr 2024 – Present

Laboratory of Brain-Computer Interfaces and Machine Learning, HUST

- Continuous-state EEG applications such as drowsiness and attention monitoring require transfer across users, but most class-conditional methods assume discrete labels. Proposed **C2R** to represent continuous targets with fuzzy class memberships, extending classification-based transfer algorithms to EEG regression and improving generalization across users and continuous decoding tasks. *(IEEE TETCI, Early Access, 2026)*
- EEG decoders deployed to a new user face simultaneous shifts in signal distributions, signal–behavior relationships, and personal label scales. Proposed **MCL-Align** to align feature distributions, conditional directions, and label statistics, enabling more reliable plug-and-play regression with modest training overhead. *(IEEE TBME, under review)*
- Streaming or embedded EEG interfaces cannot afford backpropagation whenever recording conditions change. As equal-first author, co-developed **BFT**, a forward-only test-time adaptation method that aggregates reliability-aware transformed predictions, enabling efficient adaptation for resource-constrained and quantized neural interfaces. *(IEEE JBHI, accepted, 2026)*

Cross-Subject Brain-to-Image Retrieval
Dec 2025 – Present

- Brain-to-image retrieval for a new user often fails because neural representations preserve semantics in different subject-specific coordinate systems. Diagnosed this mismatch and proposed **SCORE** to recover an unseen user's coordinates through landmark matching and an orthogonal map, enabling label-free transfer without target labels, source EEG, or encoder updates. *(manuscript under review)*
- Practical neural retrieval must work from one or a few recordings rather than trial averages. Identified non-transitive neural–visual alignment as the key failure mode and proposed **NEAR** to anchor both query and gallery representations, improving few-repetition retrieval consistently across EEG, MEG, and fMRI. *(manuscript under review)*

Selected Projects and Competitions

Fast and Accurate EEG-Controlled Robotic Arm
Sep 2025 – Dec 2025

- Addressed reliable, low-latency intention decoding in a live competition setting by designing the EEG paradigm, decoder, and closed-loop robotic control system. The resulting interface won the **national championship** at the 2025 China Brain-Computer Interface Competition and was deployed as a demonstration system for Wuhan Optics Valley Biotech City; covered by CCTV, *Global Times*, and *Shanghai Science & Technology*.

ECoG-Based Limb-Movement Intention Decoding
2024

- Developed an ECoG-based decoder for continuous limb-movement intentions and won **First Prize** in the minimally invasive BCI track at the 2024 World Robot Contest Championship.

Industry Experience

Alibaba Cloud Intelligence Group
Jul 2026 – Present

Research Intern

- Cloud-failure prediction is hindered by noisy, heterogeneous log templates and irregular event timing. Built a domain-general tokenizer and pretrained log encoder with masked-language and contrastive objectives, then combined TAAT's pairwise time relations with RoPE so the classifier models elapsed time and event order separately.
- To reduce repetitive manual experimentation, developed an internal **AutoResearch** framework that closes the loop from hypothesis generation and controlled training to metric evaluation and selection of the next experiment.

Dongfeng Motor Group — Multimodal Intelligent Cockpit Evaluation
Jun 2025 – Jun 2026

Industry-funded project — Project Lead

- Addressed the lack of an objective way to evaluate in-vehicle user experience from heterogeneous human signals. Led the project from protocol and corpus design to multimodal modeling, integrating EEG, speech, and facial cues into a unified scoring, visualization, and diagnostic system delivered to Dongfeng Motor Group.

Selected Awards and Honors

- **National Champion**, China Brain-Computer Interface Competition, Brain-Controlled Robotic Arm track, 2025.
- **First Prize**, ECoG-Based Limb-Movement Intention Decoding, World Robot Contest Championship, 2024.
- **First Prize**, National Undergraduate Intelligent Vehicle Competition, Multi-Vehicle Formation track, 2022.
- First-Class Graduate Academic Scholarship, HUST, 2024–25 and 2025–26; Yunnan Scholarship, 2024–25; Huawei Scholarship, 2021–22.
- Outstanding Graduate of HUST, 2024.